

Theorem A: initialized escape and continuum continuation

Working proof document

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Status and source contract

Theorem A is not yet proved. This document establishes new analytic, initialized early-phase estimates, including a quantitative mass-escape statement at the canonical parameters. It also proves a continuous-label reference construction, its interpolation-defect estimates, and explicit state-to-observable bounds for a subsequent computer-assisted continuation. The reference construction is not itself a completed trajectory certificate.

The original theory files are unchanged. The principal inputs are `RELU_POPULATION_FOUNDATIONS_v2.tex` (F), `THEORY_HANDOFF_THEOREMS_A_B.md` (H), and `LEAK_COMPENSATION_FACTORIZATION_AUDIT.md` (LC). In particular, use F P1–P7 for the characteristic solution and exact observables, F P12 for the signed second variation, and H Section 4 for the intended final Theorem A. No statement below imports a learned-state entry assumption from the older appendix. No repository content was fetched or changed for this working document.

Chosen route for A: certify the exact cluster rates directly. Completion of the reciprocal-compensation factorization in Theorem B is not a prerequisite. All estimates below concern the full population, not a fitted two-block system.

Notation

All derivatives in this document use **normalized time** $\tau = \mu_1^2 t$, where t is physical time. Set

$$P_1 = \mathcal{N}(e_1, q^2 I_2), \quad P_2 = \mathcal{N}(\rho e_2, q^2 I_2), \quad P = (P_1 + P_2)/2,$$

$$A = \mathbb{E}_P[XX^\top] = \text{diag}(1/2 + q^2, \rho^2/2 + q^2), \quad a_* := \|A\| = 1/2 + q^2.$$

The label law is $\lambda(d\alpha) = d\alpha/(2\pi)$. The initialized, balanced characteristic solution is

$$U = \sqrt{m} a_\theta, \quad W = \sqrt{m} b_\psi, \quad S = \int m d\lambda,$$

$$U_0(\alpha) = W_0(\alpha) = \sqrt{S_0} a_\alpha, \quad f(x) = \int W(U^\top x)_+ d\lambda.$$

Define

$$A_\theta = \mathbb{E}_P[XX^\top \mathbf{1}_{\{a_\theta^\top X > 0\}}], \quad C_f(\theta) = \mathbb{E}_P[f(X)X^\top \mathbf{1}_{\{a_\theta^\top X > 0\}}].$$

Thus $R_f = A_\theta - C_f$, and the exact equations are

$$U' = R_f^\top W, \quad W' = R_f U,$$

$$\theta' = b^\top R_f a^\perp, \quad \psi' = (b^\perp)^\top R_f a, \quad (\log m)' = 2b^\top R_f a.$$

The reference objects introduced below are distinguished by hats. In the early-phase comparison, the hat denotes the exact, decoupled **target-only flow**. In the later numerical-interface sections, it denotes the **polygonally interpolated reference**. These are different references and must not be identified.

1. R3a — initialized spectral and alignment bounds

Status: PROVED, from the specified initialization.

Proposition 1

Let

$$\bar{S}(\tau) = S_0 e^{2a_* \tau}, \quad u(\tau) = \bar{S}(\tau) - S_0,$$

and

$$I(\tau) = a_* \int_0^\tau \varepsilon_{\text{al}}(s)^2 ds = S_0^2 \left[\frac{e^{4a_* \tau} - 1}{4} - (e^{2a_* \tau} - 1) + a_* \tau \right].$$

For the exact initialized population solution,

$$S(\tau) \leq \bar{S}(\tau) \quad (\tau \geq 0).$$

On any interval on which $\varepsilon_{\text{al}}(\tau) < \pi$, choose continuous angular lifts with $\psi_0 - \theta_0 = 0$. Then, uniformly in the initial label,

$$\boxed{|\psi_\tau - \theta_\tau| \leq \varepsilon_{\text{al}}(\tau),}$$

$$\boxed{S_0 e^{-\varepsilon_{\text{al}}(\tau) - I(\tau)} \leq m_\tau(\alpha) \leq S_0 e^{2a_* \tau + \varepsilon_{\text{al}}(\tau).}$$

These bounds include a positive input-radius guard. They do not assume persistent vector alignment.

Proof

First,

$$0 \preceq A_\theta \preceq A \preceq a_* I.$$

A useful improvement over the bound involving the full second-moment trace is

$$\boxed{\|C_f(\theta)\| \leq a_* S.}$$

Indeed, for unit vectors v, z , substitute the balanced feature representation of f to obtain

$$|v^\top C_f(\theta) z| \leq \int m(\alpha) \mathbb{E} \left[(a_{\theta(\alpha)}^\top X)_+ |z^\top X| \mathbf{1}_{\{a_\theta^\top X > 0\}} \right] d\lambda.$$

Cauchy–Schwarz bounds each expectation by

$$\sqrt{\mathbb{E}(a_{\theta(\alpha)}^\top X)^2} \sqrt{\mathbb{E}(z^\top X)^2} \leq a_*.$$

This proves the claim. The same argument also gives $\|f\|_{L^2(P)} \leq \sqrt{a_*} S$.

The exact mass identity from F P5 now yields

$$S' = 2\mathbb{E}[X^\top f] - 2\|f\|_{L^2(P)}^2 \leq 2a_* S,$$

because

$$\mathbb{E}[X^\top f] = \int m b^\top A_\theta a d\lambda \leq a_* S.$$

Multiplying the differential inequality by $e^{-2a_* \tau}$ proves the mass envelope.

Set $\delta = \psi - \theta$. For any symmetric planar matrix B ,

$$(b^\perp)^\top B a - b^\top B a^\perp = -\text{tr}(B) \sin \delta.$$

Consequently the actual alignment defect satisfies

$$\delta' = -\text{tr}(A_\theta) \sin \delta + \zeta,$$

where

$$\zeta = -(b^\perp)^\top C_f a + b^\top C_f a^\perp, \quad |\zeta| \leq 2a_* S.$$

For $|\delta| < \pi$, the target term points toward zero, so

$$D^+ |\delta| \leq 2a_* S \leq 2a_* \bar{S}.$$

Integration gives $|\delta| \leq \varepsilon_{\text{al}}$. A first exit from $(-\pi, \pi)$ would contradict $\varepsilon_{\text{al}} < \pi$, closing this guard from initialization.

For the radial estimates, put $g = b^\top R_f a$. The upper bound is

$$g \leq a_* + a_* S.$$

For the lower bound, positivity of A_θ gives

$$b^\top A_\theta a = \frac{1}{2} \left(a^\top A_\theta a + b^\top A_\theta b - (b - a)^\top A_\theta (b - a) \right) \geq -\frac{a_*}{2} |b - a|^2 \geq -\frac{a_*}{2} \delta^2.$$

Thus

$$-a_* \delta^2 - 2a_* S \leq (\log m)' \leq 2a_* + 2a_* S.$$

Integrate and use $2a_* \int_0^\tau \bar{S} = \varepsilon_{\text{al}}(\tau)$ and $a_* \int_0^\tau \varepsilon_{\text{al}}^2 = I(\tau)$. This proves all assertions. $\square$

2. R3b — comparison with the exact target-only flow

Status: PROVED. This is a comparison theorem, not a replacement of the nonlinear residual by the target.

For each initial label define an independent reference neuron V by

$$V' = \mathbb{E}_P[X(V^\top X)_+], \quad V_0 = \sqrt{S_0} a_\alpha.$$

Set $\hat{U} = \hat{W} = V$, $\hat{m} = |V|^2$, and let $\hat{\theta}$ be its angle. Its input and output vectors agree because the target-only residual matrix is symmetric. This assertion is about the reference only.

Write

$$v_0(\theta) = (a_\theta^\perp)^\top A_\theta a_\theta, \quad g_0(\theta) = a_\theta^\top A_\theta a_\theta.$$

The reference equations are

$$\hat{\theta}' = v_0(\hat{\theta}), \quad (\log \hat{m})' = 2g_0(\hat{\theta}).$$

Proposition 2

Define

$$H(\tau) = S_0 \left(2e^{2a_*\tau} - 3e^{a_*\tau} + 1 \right).$$

On any interval where $\varepsilon_{\text{al}}(\tau) < \pi$, the exact initialized flow and the target-only reference satisfy, labelwise,

$$\boxed{|\theta - \hat{\theta}| \leq H, \quad |\psi - \hat{\theta}| \leq H + \varepsilon_{\text{al}},}$$

$$\boxed{\left| \log \frac{m}{\hat{m}} \right| \leq 2H.}$$

Proof

The moving-boundary formula implies $A'_\theta a_\theta = 0$. Therefore

$$v'_0(\theta) = (a_\theta^\perp)^\top A_\theta a_\theta^\perp - a_\theta^\top A_\theta a_\theta, \quad |v'_0| \leq a_*.$$

Also,

$$g'_0(\theta) = 2(a_\theta^\perp)^\top A_\theta a_\theta, \quad |g'_0| \leq a_*.$$

For the latter inequality, evaluate the quadratic form of A_θ on $(a_\theta + a_\theta^\perp)/\sqrt{2}$ and $(a_\theta - a_\theta^\perp)/\sqrt{2}$: both values lie in $[0, a_*]$, and their difference is $2(a_\theta^\perp)^\top A_\theta a_\theta$.

The true input velocity obeys

$$|\theta' - v_0(\theta)| \leq a_*|b - a| + a_*S \leq a_*(\varepsilon_{\text{al}} + \bar{S}).$$

Thus $h = |\theta - \hat{\theta}|$ satisfies

$$D^+h \leq a_*h + a_*(\varepsilon_{\text{al}} + \bar{S}), \quad h(0) = 0.$$

The displayed H solves this scalar inhomogeneous equation exactly, so $h \leq H$.

Similarly,

$$|b^\top R_f a - g_0(\hat{\theta})| \leq a_*(\varepsilon_{\text{al}} + \bar{S} + H) = H'.$$

Integrating the logarithmic mass equations gives $|\log(m/\hat{m})| \leq 2H$. The output-angle estimate follows by adding Proposition 1's alignment defect. $\square$

Useful feature of the estimate. The constants in these early angular comparisons do not contain the supremum of the Gaussian angular density or a factor $1/q$. The gate derivative cancels because it is multiplied by the input direction. The bound still grows with time and is not being extrapolated to the learned reversal window.

3. R3c — an explicit initialized escape at the canonical parameters

Status: PROVED. This is a mass-escape result, not substantial learning in the sense of Theorem A(i).

Take

$$\rho = 2/3, \quad q = 1/20, \quad S_0 = 1/5000, \quad a_* = 201/400,$$

and fix the initial-label set

$$C_0 = [-\pi/12, \pi/12] \quad \text{modulo } 2\pi.$$

Its label probability is $1/12$. It is defined at initialization, not selected after observing the trajectory.

Proposition 3

For every $0 \leq \tau \leq 6$, uniformly in all labels,

$$S(\tau) \leq 0.0832, \quad |\psi - \theta| \leq 0.0830,$$

$$1.83 \times 10^{-4} \leq m_\tau(\alpha) \leq 0.091, \quad |U_\tau(\alpha)| = |W_\tau(\alpha)| > 0.0135.$$

Moreover,

$$\boxed{S(6) > 16S_0 = 0.0032.}$$

For $\mu_1 = 3$, the endpoint is physical time $t = 6/9 = 2/3$.

Proof of the reference sector guard

Let $\theta_0 = \pi/12$, and write $c = \cos \theta$, $s = \sin \theta$. The Gaussian feature formula in F P4 gives

$$v_0(\theta) = \frac{1}{2} [-s\{c\Phi(c/q) + q\phi(c/q)\} + \rho c\{\rho s\Phi(\rho s/q) + q\phi(\rho s/q)\}].$$

At $\theta = \theta_0$, use $\Phi(c/q) \geq 1/2$, $\Phi(\rho s/q) \leq 1$, and $\phi \leq 2/5$. It follows that

$$v_0(\theta_0) \leq \frac{c}{2} \left[-\frac{s}{18} + \frac{1}{75} \right] < 0,$$

since $s = \sin(\pi/12) > 1/4$ and $1/72 > 1/75$.

At $\theta = -\theta_0$, both terms in the original displayed formula are strictly positive: $-s > 0$, $c > 0$, and each bracket is the expectation of the positive part of a nondegenerate Gaussian. Therefore $v_0(-\theta_0) > 0$. The interval $[-\theta_0, \theta_0]$ is invariant for the reference scalar ODE.

For a reference label in this interval,

$$g_0(\hat{\theta}) = \mathbb{E}_P[(a_{\hat{\theta}}^\top X)_+]^2 \geq \frac{1}{2} \cos^2 \theta_0.$$

Indeed, on cluster 1 the projected Gaussian has positive mean $\cos \hat{\theta}$, and $\mathbb{E}[Y_+^2] \geq (\mathbb{E}Y_+)^2 \geq (\mathbb{E}Y)^2$. Consequently

$$\hat{m}_\tau(\alpha) \geq S_0 e^{\cos^2 \theta_0 \tau} \quad (\alpha \in C_0),$$

and Proposition 2 gives the full-population lower bound

$$S(\tau) \geq \int_{C_0} m_\tau d\lambda \geq \frac{S_0}{12} \exp\{\cos^2(\pi/12)\tau - 2H(\tau)\}.$$

Quantitative arithmetic and closure

The elementary exponential bounds

$$e^{6.03} < 416, \quad e^{3.015} > 20$$

imply, using monotonicity of $\bar{S}$, ε_{al} , and H , throughout $[0, 6]$,

$$\bar{S} \leq 0.0832, \quad \varepsilon_{\text{al}} \leq 0.0830 < \pi, \quad H \leq 0.1546.$$

The bound on ε_{al} closes Proposition 1's angular guard over the whole interval. Also,

$$I(6) \leq S_0^2(416^2/4 + 3.015) = 0.0017306806.$$

Using $e^{-x} \geq 1 - x$ in the lower radial estimate yields $m > 1.83 \times 10^{-4}$. Using $e^x \leq 1/(1 - x)$ for $0 \leq x < 1$ in the upper estimate yields $m \leq 0.0832/(1 - 0.0830) < 0.091$. Squaring 0.0135 verifies the asserted radius guard.

Finally,

$$6 \cos^2(\pi/12) - 2H(6) > 6(0.933) - 0.3092 = 5.2888,$$

and $e^{5.2888} > 192$. Thus $S(6) > (S_0/12)192 = 16S_0$.

All decimal inequalities used here can be checked with rational Taylor bounds; `verify_theorem_A_escape_bounds.py` implements those checks using exact rational arithmetic. It does not integrate a training trajectory. $\square$

What this does and does not establish

The proof gives actual growth from the isotropic initialization, control of the true angular mismatch, a labelwise radius lower bound, and a controlled reference comparison. It does not assume that a strong family has already formed. It does not prove that the weak concept is substantially learned, that either late cluster rate has a prescribed sign, or that the trajectory has entered the competition region. The simple envelope should not be used through physical time 3.5–4.

4. R2a — a continuous-label numerical reference with quadratic initialization error

Status: PROVED construction and estimates. Application to a numerical trajectory remains open.

The numerical continuation should not be identified with the continuum merely because the label grid is fine. The following explicit lift gives a continuous labeled reference to which F P12 applies.

Let $L \geq 3$, $\alpha_j = 2\pi j/L$, with periodic indexing. Let $Z_j(\tau) = (U_j(\tau), W_j(\tau))$ be continuous, piecewise C^1 nodal trajectories. On the cell $\alpha = \alpha_j + 2\pi s/L$, $0 \leq s \leq 1$, define

$$\hat{Z}(\tau, \alpha) = (1 - s)Z_j(\tau) + sZ_{j+1}(\tau).$$

This is Cartesian interpolation. It need not preserve balance or remain on the input/output unit circles. That is allowed: use the non-balanced version of F P12.

The Hilbert norm is

$$\|Z\|_{\mathcal{H}}^2 = \int (|U|^2 + |W|^2) d\lambda.$$

Proposition 4 — initialization enclosure

Suppose the ideal nodal initialization is exact:

$$Z_j(0) = (\sqrt{S_0}a_{\alpha_j}, \sqrt{S_0}a_{\alpha_j}).$$

Then

$$\|Z_0 - \hat{Z}_0\|_{\mathcal{H}} \leq \|Z_0 - \hat{Z}_0\|_{L^\infty_\alpha} \leq \frac{\sqrt{2S_0}\pi^2}{2L^2}.$$

Also,

$$\inf_{\alpha} |\hat{U}_0(\alpha)| = \sqrt{S_0} \cos(\pi/L) > 0.$$

At the canonical mass and $L = 16384$, the initial error has the rigorous bound

$$r_0, u_0 \leq \frac{1}{10L^2} < 3.73 \times 10^{-10}.$$

The constant uses $\sqrt{2S_0} = 1/50$ and $\pi^2 < 10$. This is an initialization error, not a bound on the error at the learned window.

Proof

For a twice differentiable vector-valued curve g on a cell of length h , the norm of the difference from its endpoint linear interpolant is at most $h^2 \sup |g''|/8$. Apply this to $g(\alpha) = (\sqrt{S_0}a_\alpha, \sqrt{S_0}a_\alpha)$, for which $|g''| = \sqrt{2S_0}$ and $h = 2\pi/L$. The Hilbert norm is bounded by the supremum norm because λ is a probability measure. The minimum radius of a circular chord occurs at its midpoint. $\square$

Floating-point node initialization. If the stored nodal values differ from the ideal trigonometric initialization by at most ϵ_{node} in the joint Cartesian norm, add ϵ_{node} to both initial error bounds. Initial-mass or parameter perturbations likewise require their own explicit enclosure; they are not absorbed automatically.

Exact label integration of the reference output

For a fixed input x , define on cell j

$$u_j = U_j^\top x, \quad v_j = (U_{j+1} - U_j)^\top x, \quad \Delta W_j = W_{j+1} - W_j.$$

On the subinterval where $u_j + sv_j > 0$, the feature integrand is

$$(W_j + s\Delta W_j)(u_j + sv_j).$$

Its vector-valued primitive is

$$\mathcal{P}_j(s) = W_j u_j s + \frac{W_j v_j + \Delta W_j u_j}{2} s^2 + \frac{\Delta W_j v_j}{3} s^3.$$

There is at most one crossing per cell. Thus

$$\hat{f}(x) = \frac{1}{L} \sum_j [\mathcal{P}_j(s_{j,+}) - \mathcal{P}_j(s_{j,-})],$$

with an empty active interval contributing zero. There is no label quadrature error in this mathematical formula. Interval evaluation still has to enclose arithmetic and any uncertain branch. Gaussian integration, nodal time interpolation, and integration of the trajectory are separate obligations.

5. R2b — an explicit reference-defect bound

Status: PROVED estimate. Its numerical inputs have not been supplied or evaluated here.

At a fixed normalized time, hold $\hat{f}$ fixed and define the single-label field

$$\mathcal{F}_{\hat{f}}(U, W) = \left(R_{\hat{f}}(\arg U)^\top W, R_{\hat{f}}(\arg U)U \right).$$

The nodal residual must be evaluated against the lifted continuum output:

$$d_j = Z'_j - \mathcal{F}_{\hat{f}}(Z_j).$$

It is not sufficient to use a residual against the original particle quadrature output.

On cell j , let

$$r_j = \min_{0 \leq s \leq 1} |U_j + s\Delta U_j| > 0, \quad w_j = \max(|W_j|, |W_{j+1}|).$$

The radius minimum is exactly computable by minimizing a quadratic over $[0, 1]$. Suppose J bounds $\|\partial_\theta R_{\hat{f}}\|$ and K bounds the Lipschitz constant of $\partial_\theta R_{\hat{f}}$ on the relevant angles. Put

$$C_j = \frac{2J}{r_j} + \frac{(K + 2J)w_j}{r_j^2}.$$

Proposition 5

For the true continuum defect

$$\mathfrak{d} = \widehat{Z}' - F(\widehat{Z}),$$

one has

$$\|\mathfrak{d}\|_{\mathcal{H}} \leq \left(\frac{1}{L} \sum_j |d_j|^2 \right)^{1/2} + \left(\frac{1}{120L} \sum_j C_j^2 |\Delta Z_j|^4 \right)^{1/2},$$

$$\|\mathfrak{d}\|_{\infty} \leq \max_j |d_j| + \frac{1}{8} \max_j C_j |\Delta Z_j|^2.$$

Proof

Consider a Cartesian line $(U, W) + s(v, z)$ with input radius bounded below by r_j . Write $a = U/|U|$, $n = a^\perp$, and $\theta = \arg U$. Then

$$|\theta'| \leq |v|/r_j, \quad |\theta''| \leq 2|v|^2/r_j^2.$$

The identity $R'(\theta)a = 0$ holds for every continuous homogeneous residual. It implies

$$\frac{d}{ds}[R(\theta)U] = Rv, \quad \left| \frac{d^2}{ds^2}[R(\theta)U] \right| \leq (J/r_j)|v|^2.$$

For the other component,

$$\frac{d^2}{ds^2}[R(\theta)^\top W] = 2R'^\top z\theta' + R''^\top W(\theta')^2 + R'^\top W\theta''.$$

Using $2|v||z| \leq |v|^2 + |z|^2$, and bounding the norm of the pair by the sum of its component norms, gives

$$\left| \frac{d^2}{ds^2} \mathcal{F}_{\hat{f}}((U, W) + s(v, z)) \right| \leq C_j(|v|^2 + |z|^2).$$

Lipschitz first derivatives suffice; the second derivative may be interpreted almost everywhere. The linear-interpolation remainder on the unit cell is consequently at most

$$\frac{C_j}{2} s(1-s) |\Delta Z_j|^2.$$

The defect equals this field-interpolation remainder plus the linear interpolant of d_j, d_{j+1} . Jensen's inequality bounds the latter's global squared Hilbert norm by $L^{-1} \sum_j |d_j|^2$. For the former, use $\int_0^1 s^2(1-s)^2 ds = 1/30$. Minkowski's inequality and $\max s(1-s) = 1/4$ give the two displayed conclusions. $\square$

Explicit but conservative J, K

Let B be a Lipschitz bound for $\hat{f}$, for example

$$B = \frac{1}{2} \|\hat{Z}\|_{\mathcal{H}}^2.$$

For the mixture angular weight ω in F P4 put $\Omega_0 = \|\omega\|_{\infty}$, $\Omega_1 = \|\omega'\|_{\infty}$. Valid constants are

$$J = 2\Omega_0(1+B), \quad K = 2(\Omega_1 + 2\Omega_0)(1+B).$$

To verify this, write

$$R'(\theta) = T(\theta)n^{\top},$$

$$T(\theta) = \omega(\theta + \pi/2)e_{\hat{f}}(n) + \omega(\theta - \pi/2)e_{\hat{f}}(-n).$$

The unit-circle residual has supremum and Lipschitz constant at most $1+B$. Thus $|T| \leq J$ and $\text{Lip}(T) \leq 2(\Omega_1 + \Omega_0)(1+B)$; differentiating n adds another J . Local validated residual bounds can be much sharper than these global envelopes. No claim is made that the conservative constants already fit the late-time budget.

6. Closing the two compatible error radii

Status: PROVED comparison interface; the required reference enclosures remain OPEN.

A Hilbert error alone does not establish labelwise input-radius or probe-gate guards. The polygonal reference gives a way to propagate both norms.

Let

$$r = \|Z - \hat{Z}\|_{\mathcal{H}}, \quad u_{\infty} = \|Z - \hat{Z}\|_{\infty}, \quad N = \|\hat{Z}\|_{\mathcal{H}}, \quad R_{\infty} = \|\hat{Z}\|_{\infty}.$$

The output-difference estimate is

$$\|f_Z - \hat{f}\|_{L^2(P)} \leq \sqrt{a_*}(N + r)r.$$

Suppose $\ell(\tau, r, u)$ bounds the full-flow one-sided quadratic form from F P12 on every reference-to-candidate segment, and $\ell_0(\tau, r, u)$ bounds the local single-label one-sided form for $\mathcal{F}_{\hat{f}}$ on those segments. Then

$$D^+r \leq \ell(\tau, r, u)r + \|\mathfrak{d}\|_{\mathcal{H}},$$

$$D^+\|Z - \hat{Z}\|_{\infty} \leq \ell_0(\tau, r, u)u + a_*(R_{\infty} + u)(N + r)r + \|\mathfrak{d}\|_{\infty}$$

when the actual supremum error is u .

For the second inequality, split the difference of fields into a same-residual local change and the effect of replacing $\hat{f}$ by f_Z . The latter, at a label with state (U, W) , has joint norm at most

$$\sqrt{a_*}\sqrt{|U|^2 + |W|^2} \|f_Z - \hat{f}\|_{L^2(P)}.$$

The same-residual part is controlled by ℓ_0 . Its Jacobian is the local residual-curvature matrix in F P12, without the nonlocal negative Gauss–Newton term. It uses the full ratio $W/|U|$, not a balanced surrogate.

Positive strict supersolutions for both scalar inequalities, initialized above the Proposition 4 bounds, close a first-exit argument provided

$$u_{\text{envelope}}(\tau) < \inf_{\alpha} |\hat{U}(\tau, \alpha)|,$$

where u_{envelope} denotes the supremum-error envelope. Every segment then has input radius at least $\inf |\hat{U}| - u_{\text{envelope}} > 0$.

The bound on ℓ must hold for the candidate *full state*, including its output residual. The local bound ℓ_0 holds with $\hat{f}$ frozen. They cannot be interchanged without proof. A reference-only eigenvalue maximum is not a tube bound.

7. R2c — exact reference strips and explicit observable errors

Status: PROVED static interfaces; late-window values remain OPEN.

7.1 Probe-strip mass is computable cell by cell

For a unit probe ξ and a supremum Cartesian error envelope u , define

$$\hat{\Gamma}_W(\xi, u) = \int |\widehat{W}(\alpha)|^2 \mathbf{1}_{\{|\widehat{U}(\alpha)^{\top} \xi| \leq u\}} d\lambda.$$

Every mismatch between the true and reference probe indicators lies in this set. On a cell, $\widehat{U}^{\top} \xi$ is affine in s , and $|\widehat{W}|^2$ is quadratic. Thus its contribution is exactly obtained by integrating

$$|W_j|^2 + 2sW_j^\top \Delta W_j + s^2|\Delta W_j|^2$$

over the subinterval satisfying $|U_j^\top \xi + s\Delta U_j^\top \xi| \leq u$.

Both probe boundaries are included automatically. No bounded projected-density hypothesis, global monotonicity of the input-angle map, or exclusion of folds is needed for this finite-tolerance calculation. A whole cell exactly on a probe boundary contributes its full weight and is not silently removed. Uniform probe-sector evaluation still needs interval treatment of the probe angle.

7.2 Learning observables

Let r, u enclose the two state errors and set

$$\epsilon_f = (N + r)r.$$

Then

$$\sup_{|x|=1} |f_Z(x) - \hat{f}(x)| \leq \epsilon_f, \quad \|f_Z - \hat{f}\|_{L^2(P_p)} \leq \sqrt{Q_p} \epsilon_f,$$

where $Q_1 = 1 + q^2$ and $Q_2 = \rho^2 + q^2$.

Indeed, split the output difference as

$$\int \Delta W(U^\top x)_+ d\lambda + \int \widehat{W}\{(U^\top x)_+ - (\widehat{U}^\top x)_+\} d\lambda$$

and apply Cauchy–Schwarz in labels. For the training L^2 estimate, use $\|\mathbb{E}_{P_p} X X^\top\| = Q_p$ before applying the same label bound.

Consequently the partition-free learning observables satisfy

$$\boxed{|\mathfrak{m}_p - \widehat{\mathfrak{m}}_p| \leq \epsilon_f,}$$

$$\boxed{\mathcal{L}_p \leq \frac{1}{2} \left(\sqrt{2\widehat{\mathcal{L}}_p} + \sqrt{Q_p} \epsilon_f \right)^2.}$$

The first bound follows by pairing the p -th output difference with X_p and dividing by Q_p . Neither inequality supplies a signed selected residual moment.

7.3 Exact cluster-rate error

For any, possibly unbalanced, Cartesian state define

$$V_p(Z; \xi) = \int \left[(U^\top \xi)_+ R_{p,Z}(\arg U) U + \mathbf{1}_{\{U^\top \xi > 0\}} W W^\top R_{p,Z}(\arg U) \xi \right] d\lambda.$$

The physical rate is

$$D_p(Z; \xi) = -k e_Z(\xi)^\top V_p(Z; \xi), \quad k = \mu_1^2/2.$$

Assume $u < r_{\min} := \inf |\widehat{U}|$ and put

$$\eta_\theta = \arcsin(u/r_{\min}).$$

Let J_p bound $\|\partial_\theta R_{p,\widehat{f}}\|$ on the angle segments between matched inputs. Then

$$\|R_{p,Z}(\arg U) - R_{p,\widehat{f}}(\arg \widehat{U})\| \leq \Delta R_p := Q_p \epsilon_f + J_p \eta_\theta.$$

The same-angle term uses Cauchy–Schwarz and the preceding training L^2 bound. The angle term uses the reference residual derivative. The angular estimate is the tangent-angle bound for a ball of radius u around a nonzero vector of norm at least $r_{\min}$.

Let H_p bound both residual matrices at matched inputs. Define

$$\mathcal{V}_p = H_p(2N + r)r + N^2 \Delta R_p + H_p \widehat{\Gamma}_W(\xi, u).$$

Then, for unit probes,

$$|V_p(Z; \xi) - V_p(\widehat{Z}; \xi)| \leq \mathcal{V}_p,$$

$$|D_p(Z; \xi) - D_p(\widehat{Z}; \xi)| \leq k \left[(|e_{\widehat{f}}(\xi)| + \epsilon_f) \mathcal{V}_p + \epsilon_f |V_p(\widehat{Z}; \xi)| \right].$$

Proof of the velocity bound

For the first integrand, hold the true residual matrix fixed while changing U , and then change the matrix. Its difference is bounded by

$$H_p(|U| + |\widehat{U}|) |\Delta U| + \Delta R_p |\widehat{U}|^2.$$

For the second integrand, first hold its indicator fixed. Changing W and the matrix costs

$$H_p(|W| + |\widehat{W}|) |\Delta W| + \Delta R_p |\widehat{W}|^2.$$

The remaining indicator difference is bounded by $H_p |\widehat{W}|^2$ on the reference strip. Integrating and applying Cauchy–Schwarz to the two smooth parameter terms gives $H_p(\|Z\|_{\mathcal{H}} + N)r \leq H_p(2N + r)r$. The residual-matrix terms sum to $N^2 \Delta R_p$. This proves the first bound. For the rate bound, subtract the two scalar contractions and use $|e_Z - e_{\widehat{f}}| \leq \epsilon_f$. $\square$

Quantitative choice of constants. Near the learned window, do not automatically use $H_p = M_{2,p}(1 + S)$. A valid sharper alternative is $\|R_{p,Z}\| \leq \sqrt{2\mathcal{L}_p Q_p}$, or a direct validated angular evaluation. The derivative constant J_p can also use the actual weighted boundary residual from F P4. Whether these sharper bounds close the budget must be computed, not presumed.

8. Central-witness certificate contract for Theorem A

The following numerical choices are **design targets**, not proved late-time bounds:

- Physical learned interval $J = [3.45, 4.05]$, with $J_- = [3.45, 3.50]$ and $J_+ = [4.00, 4.05]$.
- Substantial learning: $m_p \geq 1/2$ and $\mathcal{L}_p \leq \mathcal{L}_p(0)/2$ on J .
- Competition: $D_1 \leq -5 \times 10^{-4}$ and $D_2 \geq 5 \times 10^{-4}$ on J .
- Early/late total-rate margins $\gamma_- = \gamma_+ = 3 \times 10^{-4}$.
- A provisional per-cluster rate-error allowance 10^{-4} , including every continuum, time, integration, and arithmetic error. Its allocation must be derived from the preceding inequalities.

At the central probe and parameter, H reports the floating-point totals $-7.9276072 \times 10^{-4}$ at $t = 3.5$ and $+8.4101882 \times 10^{-4}$ at $t = 4$. These motivate separated witness windows. They do not certify any part of J or any nearby parameter/probe.

The first validated reference may be continued directly from zero, using Proposition 4, or may combine an early analytic phase with a later numerical phase. In the latter case, the actual interface must bound the distance to the chosen numerical reference; the mass-escape lower bound alone is not a sufficient entry enclosure.

After a central certificate closes, certify a positive-width probe sector containing -85° and a parameter box containing $(2/3, 0.05, 2 \times 10^{-4})$ in its interior. No sector or box width is certified here. A common fixed physical interval avoids invoking an unproved hitting-time clock; μ_1 remains fixed when varying the normalized parameter triple.

If the above target margins eventually hold, absolute continuity gives an interior minimizer t_* and

$$E_\xi(3.45) - E_\xi(t_*) \geq 1.5 \times 10^{-5}, \quad E_\xi(4.05) - E_\xi(t_*) \geq 1.5 \times 10^{-5}.$$

These are conditional arithmetic consequences of the proposed window widths and margins, **not currently established reversals**.

9. Proof ledger and next quantitative obligation

Item	Status after this work
Spectral target/remainder bounds and initialized angle/radius guards	Proved in Proposition 1
Uniform comparison with an exact target-only flow	Proved in Proposition 2
Canonical growth beyond $16S_0$ by physical time $2/3$	Proved in Proposition 3
Continuous-label reference with $O(L^{-2})$ initialization error	Proved in Proposition 4
Reference-defect reduction to nodal residuals and cell curvature	Proved in Proposition 5
Compatible Hilbert/supremum continuation inequalities	Proved interface; numerical hypotheses open
Exact reference strip integrals and learning/rate error interfaces	Proved static estimates

Item	Status after this work
Initialized reachability of the learned interval	Open
Both concepts substantially learned on that interval	Open
Uniform exact cluster signs and dominance switch	Open
Positive-width sector and parameter box	Open
Theorem A	Not yet proved

The next quantitative obligation is to enclose, slab by slab through the learned window,

$$\|\mathfrak{d}\|_{\mathcal{H}}, \quad \|\mathfrak{d}\|_{\infty}, \quad \ell, \quad \ell_0, \quad \inf |\widehat{U}|,$$

and close the two error supersolutions. Those enclosures then feed the learning and exact-rate estimates in Section 7. Proving only the nodal ODE has a certain rate sign would not discharge this obligation.

The mounted inputs contain the theory sources and summarized numerical evidence, but not the actual population reference arrays or integrator. A numerical application needs the canonical `labels.npz` (or equivalent nodal states and time grid), `summary.json`, and `population_pilot.py` or its exact field/time-interpolation specification. Fine probe traces can check witness-window feasibility, but cannot replace the continuum defect. This document does not claim that these missing numerical checks were run.

10. Arithmetic verification and provenance

`verify_theorem_A_escape_bounds.py` checks the numerical inequalities in Proposition 3 and the conservative initialization constant in Proposition 4 using exact rational arithmetic. Its report explicitly distinguishes these elementary checks from a training-trajectory certificate. It is not a formal proof checker for the analytic arguments.

Source SHA-256 values:

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- H, THEORY_HANDOFF_THEOREMS_A_B.md: 4eb61c25ba5f71b00a58e3845b22d4e3f948709e61875e99f6c7c43d2cad2228.
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